

GNOMP Based Range and Velocity Estimation for Millimeter Wave Radar with Few-bit ADCs

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Abstract—Few-bit analog-to-digital converters (ADCs) provide promising solutions for future low cost sensing and communication oriented applications, but it makes accurate joint range-Doppler estimation significantly more challenging. Existing two-dimensional super-resolution methods are mainly developed for unquantized measurements, while existing few-bit estimation studies mostly focus on direction-of-arrival or one-dimensional line spectrum problems. As a result, in this paper, based on the generalized orthogonal matching pursuit (GNOMP), a two dimensional super resolution and off-grid GNOMP (2D-GNOMP) is developed, addressing the multi-targets range Doppler processing for the frequency-modulated continuous-wave (FMCW) millimeter-wave radar with few-bit ADCs. To evaluate the performance, the Cramer Rao bound (CRB) is derived. 2D-GNOMP asymptotically approaches the CRB in a single and two target scenario under both 1-bit and 2-bit quantization, demonstrating near-optimal estimation performance and its potential as a practical solution for hardware-constrained FMCW radar systems.

Index Terms—Generalized Newtonized orthogonal matching pursuit, millimeter-wave radar, low resolution quantization.

I. INTRODUCTION

The FMCW millimeter-wave radar estimates the target range and velocity by transmitting and receiving linear chirp signals whose instantaneous frequency varies linearly with time [1] [2]. Owing to their high carrier frequencies, these radars offer much larger bandwidths than lower-frequency systems, inherently enabling high range resolution. Moreover, they support long coherent integration times [3], which improve detection sensitivity and estimation accuracy under low signal-to-noise (SNR) ratio conditions. Consequently, FMCW millimeter-wave radar has become increasingly important in applications such as driver assistance [4], environmental sensing [5], and human activity recognition [6].

Despite the inherent advantages of FMCW millimeter-wave radar, the use of high-speed, high-resolution ADCs results in high power consumption and hardware complexity. This has motivated increasing interest in low-resolution quantization as a practical and power-efficient solution for FMCW radar systems [7]–[9]. For example, two-dimensional super-resolution algorithms have been shown to approach single-sensor estimation-theoretic bounds for range and Doppler estimation [10]. However, it still relies on high-resolution

ADCs, which limits FMCW radar applications under stringent hardware and cost constraints.

Considering the signal distortion caused by quantization, existing works on one-bit direction of arrival (DOA) estimation established performance limits via CRB analysis [11] [12] and subsequently developed estimation algorithms [13]–[16]. In [17], line spectrum estimation and detection with few-bit ADCs were investigated, where a GNOMP algorithm with super-resolution capability was proposed. Due to its strong performance and robustness, GNOMP provides a practical framework for FMCW radar systems operating under hardware constraints, which motivates its extension to two-dimensional joint range-velocity estimation. Thus, this paper proposes a 2D-GNOMP algorithm, in which CRB is derived to evaluate performance.

Notation: Let $\mathcal{CN}(\mathbf{x}; \boldsymbol{\eta}, \boldsymbol{\Sigma})$ denote the complex normal (CN) distribution of $\mathbf{x}$ with mean $\boldsymbol{\eta}$ and covariance matrix $\boldsymbol{\Sigma}$. The standard normal probability density function (PDF) and cumulative distribution function (CDF) are defined as $\phi(x) = \exp(-x^2/2)/\sqrt{2\pi}$ and $\Phi(x) = \int_{-\infty}^x \phi(t) dt$.

II. SIGNAL MODEL

For the FMCW millimeter-wave radar employing few-bit ADCs, after down-conversion, the baseband received signal can be described as

$$y_{n,m} = \mathcal{Q}_B \left(\sum_{k=1}^K x_k e^{j2\pi(2r_k \mu T_c)n/c} e^{j2\pi(2v_k f_c T_s)m/c} + \epsilon_{n,m} \right), \quad (1)$$

$n = 0, 1, \dots, N-1$ and $m = 0, 1, \dots, M-1$, where n and m denote the index of the fast-time and slow-time domains respectively. The number of targets is K , which is supposed to be known here. Define the number of samples received G as

$$G = NM. \quad (2)$$

Let W denote the bandwidth. The Nyquist sampling is adopted, and the sampling interval T_s is $T_s = 1/W$. μ and T_c denote the chirp rate and the duration of a single chirp, $\mu = W/T_c$. f_c denotes the carrier frequency, c denotes the speed of the electromagnetic wave, $\lambda = c/f_c$ denotes the wavelength. Here we let $d = \lambda/2$. x_k , r_k and v_k denote the complex amplitude, the radial range, the radial velocity of the k th target, respectively. $\epsilon_{n,m}$ is the additive white Gaussian noise (AWGN) satisfying $\epsilon_{n,m} \sim \mathcal{CN}(0, \sigma^2)$ with σ^2 being

the variance. $\mathcal{Q}_B(\mathbf{x}) = \mathcal{Q}(\Re\{\mathbf{x}\}) + j\mathcal{Q}(\Im\{\mathbf{x}\})$, where $\mathcal{Q}(\cdot)$ is a uniform/non-uniform quantizer that maps the continuous values into a finite number of bits. Here a uniform quantizer with bit depth B is adopted [18], i.e.,

$$\mathcal{Q}_B(x) = \begin{cases} -\gamma + \frac{2\gamma}{b} \left(d + \frac{1}{2}\right), & x - d\frac{2\gamma}{b} + \gamma \in \left[0, \frac{2\gamma}{b}\right], \\ \text{sign}(x) \left(\gamma - \frac{\gamma}{b}\right), & |x| > \gamma. \end{cases} \quad (3)$$

The number of outputs of a B -bit quantizer is $b = 2^B$. γ is the maximum full-scale range, beyond which the signal will be mapped to specific extreme quantization values. Therefore, the interval size between adjacent quantization levels during the quantization process is $\frac{2\gamma}{b}$. Let $\{\tau_d\}_{d=0}^{b-1}$ be the quantizer's thresholds, the quantization interval can be expressed as $\{(\tau_d, \tau_{d+1})\}_{d=0}^{b-1}$, where $\tau_0 = -\infty$, $\tau_d = d\frac{2\gamma}{b} - \gamma$, $d = 1, 2, \dots, b-1$, $\tau_b = \infty$. To express the loglikelihood in a more compact way, two functions, $l(\mathbf{y})$ and $u(\mathbf{y})$, which represents the lower and the upper thresholds of the quantizer for a specific output of the quantizer, is adopted. For example, for the output $\frac{\gamma}{b}$ of the quantizer with bit-depth B , $l(\frac{\gamma}{b}) = 0$ and $u(\frac{\gamma}{b}) = 2\frac{\gamma}{b}$.

Define two manifold vectors

$$\mathbf{a}_N(\omega) = [1, e^{j\omega}, \dots, e^{j(N-1)\omega}]^T, \quad (4)$$

$$\mathbf{a}_M(\omega) = [1, e^{j\omega}, \dots, e^{j(M-1)\omega}]^T. \quad (5)$$

The scalar model can be formulated as the matrix-vector model described as

$$\begin{aligned} \mathbf{y} &= \mathcal{Q}_B \left(\sum_{k=1}^K x_k \mathbf{a}(\omega_{d,k}, \omega_{r,k}) + \boldsymbol{\epsilon} \right) \\ &= \mathcal{Q}_B (\mathbf{A}(\omega_d, \omega_r) \mathbf{x} + \boldsymbol{\epsilon}), \end{aligned} \quad (6)$$

where $\mathbf{y} \in \mathbb{C}^G$,

$$\mathbf{A}(\omega_d, \omega_r) = [\mathbf{a}(\omega_{d,1}, \omega_{r,1}), \dots, \mathbf{a}(\omega_{d,K}, \omega_{r,K})], \quad (7)$$

$$\mathbf{a}(\omega_{d,k}, \omega_{r,k}) = \mathbf{a}_N(\omega_r) \otimes \mathbf{a}_M(\omega_d), \quad (8)$$

ω_r is the angular frequency associated with the target range defined by $\omega_r = 2\pi f_r T_s$, where $f_r = 2r\mu/c$. ω_d is the slow-time Doppler angular frequency defined by $\omega_d = 2\pi f_d T_c$, where $f_d = 2v f_c/c$.

III. A 2D-GNOMP ALGORITHM

The main idea of the 2D-GNOMP algorithm is to estimate the multi-targets sequentially: First, estimate the first target by assuming the remaining targets are absent. Then, the first estimated target is taken into account to estimate the second target. This procedure is repeated until all K targets are estimated, at which point the algorithm terminates.

Define $\boldsymbol{\theta} = [\omega_d, \omega_r]^T$ and $\boldsymbol{\chi} = [\Re\{x\}, \Im\{x\}]^T$ as the unknown deterministic parameters. Ignoring the other targets, we try to perform a single target estimation using the following observation model

$$\mathbf{y} = \mathcal{Q}_B (\mathbf{a}(\omega_{d,1}, \omega_{r,1})x + \boldsymbol{\epsilon}). \quad (9)$$

The maximum likelihood estimates (MLE) of the range angular frequency, Doppler angular frequency and amplitude can be obtained by solving

$$(\hat{x}_{\text{ML}}, \hat{\boldsymbol{\theta}}_{\text{ML}}) = \underset{x, \boldsymbol{\theta}}{\text{argmax}} \mathcal{L}(\mathbf{y}; \boldsymbol{\theta}, x). \quad (10)$$

where the log-likelihood function is

$$\begin{aligned} \mathcal{L}(\mathbf{y}; \boldsymbol{\theta}, x) &= \sum_{g=1}^G \left[\log \left(\Phi \left(\frac{u(\Re\{y_g\}) - \Re\{a_g(\boldsymbol{\theta})x\}}{\sigma/\sqrt{2}} \right) - \Phi \left(\frac{l(\Re\{y_g\}) - \Re\{a_g(\boldsymbol{\theta})x\}}{\sigma/\sqrt{2}} \right) \right) \right. \\ &\quad \left. + \log \left(\Phi \left(\frac{u(\Im\{y_g\}) - \Im\{a_g(\boldsymbol{\theta})x\}}{\sigma/\sqrt{2}} \right) - \Phi \left(\frac{l(\Im\{y_g\}) - \Im\{a_g(\boldsymbol{\theta})x\}}{\sigma/\sqrt{2}} \right) \right) \right], \end{aligned} \quad (11)$$

Owing to the non-concave nature of the objective function with respect to both $\boldsymbol{\theta}$ and x , solving the optimization problem jointly is intractable. Nevertheless, when $\boldsymbol{\theta}$ is held fixed, the log-likelihood function becomes concave with respect to the real and imaginary components of x . This structural property motivates the adoption of an alternating minimization (AM) strategy, where the optimization procedure is initialized with a coarse estimate $(\hat{\boldsymbol{\theta}}, \hat{x})$, and subsequently refines the estimates through iterative updates.

To ensure reliable convergence, for the case where the range angular frequency ω_r , the Doppler angular frequency ω_d , and the amplitude x are all unknown, suitable initial frequency estimates can be obtained by discretizing ω_r and ω_d into finite grids, forming the range set $\mathcal{R}(\gamma_{\text{os}}^r)$ and Doppler set $\Omega(\gamma_{\text{os}}^v)$, i.e.,

$$\mathcal{R}(\gamma_{\text{os}}^r) \triangleq \left\{ n_{\text{os}} \frac{2\pi}{\gamma_{\text{os}}^r N} : n_{\text{os}} = 0, 1, 2, \dots, \gamma_{\text{os}}^r N - 1 \right\}, \quad (12)$$

$$\Omega(\gamma_{\text{os}}^v) \triangleq \left\{ m_{\text{os}} \frac{2\pi}{\gamma_{\text{os}}^v M} : m_{\text{os}} = 0, 1, 2, \dots, \gamma_{\text{os}}^v M - 1 \right\}, \quad (13)$$

where the oversampling factors γ_{os}^r and γ_{os}^v are used to construct fine search grids. Consequently, $\boldsymbol{\theta}_g$ is denoted by $\boldsymbol{\theta}_g = [\omega_{r, n_{\text{os}}}, \omega_{d, m_{\text{os}}}]^T$, where $\omega_{r, n_{\text{os}}} \in \mathcal{R}(\gamma_{\text{os}}^r)$, $\omega_{d, m_{\text{os}}} \in \Omega(\gamma_{\text{os}}^v)$. For each $\boldsymbol{\theta}_g$, we solve

$$\hat{x}_g = \underset{x}{\text{argmax}} \mathcal{L}(\mathbf{y}; \boldsymbol{\theta}_g, x). \quad (14)$$

and select

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}_g}{\text{argmax}} \mathcal{L}(\mathbf{y}; \boldsymbol{\theta}_g, \hat{x}_g). \quad (15)$$

With the coarse estimates $\hat{\boldsymbol{\theta}} = [\hat{\omega}_d, \hat{\omega}_r]^T$, a Newton-based refinement is subsequently carried out, i.e.,

$$\hat{\boldsymbol{\theta}}' = \hat{\boldsymbol{\theta}} - \left[\nabla^2 \mathcal{L}(\mathbf{y}; \hat{\boldsymbol{\theta}}, x) \right]^{-1} \bigg|_{x=\hat{x}_g} \nabla \mathcal{L}(\mathbf{y}; \hat{\boldsymbol{\theta}}, x) \bigg|_{x=\hat{x}_g}, \quad (16)$$

where $\nabla \mathcal{L}$ and $\nabla^2 \mathcal{L}$ denote the gradient and Hessian w.r.t. $\boldsymbol{\theta}$. The refined amplitude $\hat{x}'$ can be further obtained via

$$\hat{x}' = \underset{x}{\text{argmax}} \mathcal{L}(\mathbf{y}; \hat{\boldsymbol{\theta}}', x). \quad (17)$$

Denote the refined parameter estimates of the first strongest target $\hat{\boldsymbol{\theta}}'$ and $\hat{x}'$ as $\hat{\boldsymbol{\theta}}_1 = [\tilde{\omega}_{d,1}, \tilde{\omega}_{r,1}]^T$ and $\tilde{x}_1$. The 2D-GNOMP algorithm takes the strongest component estimated

into account and then iteratively estimates the second target. Accordingly, we adopt the following observation model

$$\mathbf{y} = \mathcal{Q}_B(\mathbf{a}(\tilde{\omega}_{d,1}, \tilde{\omega}_{r,1})\tilde{x}_1 + \mathbf{a}(\omega_{d,2}, \omega_{r,2})x_2 + \boldsymbol{\epsilon}). \quad (18)$$

At this stage, the effects of the strongest target on the estimation of the second target is thereby mitigated as much as possible. The log-likelihood function is turned into

$$\begin{aligned} \mathcal{L}(\mathbf{y}; \boldsymbol{\theta}, x) = & \sum_{g=1}^G \left[\log \left(\Phi \left(\frac{u(\Re\{y_g\}) - \Re\{a_g(\boldsymbol{\theta})x + \mathbf{a}(\tilde{\omega}_{d,1}, \tilde{\omega}_{r,1})\tilde{x}_1\}}{\sigma/\sqrt{2}} \right) \right. \right. \\ & - \Phi \left(\frac{l(\Re\{y_g\}) - \Re\{a_g(\boldsymbol{\theta})x + \mathbf{a}(\tilde{\omega}_{d,1}, \tilde{\omega}_{r,1})\tilde{x}_1\}}{\sigma/\sqrt{2}} \right) \Big) \\ & + \log \left(\Phi \left(\frac{u(\Im\{y_g\}) - \Im\{a_g(\boldsymbol{\theta})x + \mathbf{a}(\tilde{\omega}_{d,1}, \tilde{\omega}_{r,1})\tilde{x}_1\}}{\sigma/\sqrt{2}} \right) \right. \\ & \left. \left. - \Phi \left(\frac{l(\Im\{y_g\}) - \Im\{a_g(\boldsymbol{\theta})x + \mathbf{a}(\tilde{\omega}_{d,1}, \tilde{\omega}_{r,1})\tilde{x}_1\}}{\sigma/\sqrt{2}} \right) \right) \right]. \end{aligned} \quad (19)$$

By repeating the procedure described above, the estimate of the second target can be obtained. Fix all parameters of the first target and apply the Newton method to refine the parameter estimates of the second target. After this refinement, the parameters of the second target are fixed, and the Newton method is again employed to further refine the estimates of the first target. Finally, with the frequency estimates of both the first and second targets fixed, a joint refinement of the amplitudes is performed.

Assume that $k-1$ of the K known targets have already been estimated and refined with parameters $\{\tilde{\omega}_{r,k}, \tilde{\omega}_{d,k}, \tilde{x}_k\}_{k=1}^{k-1}$. We now perform the estimation for the k th target. The observation model can be given as

$$\mathbf{y} = \mathcal{Q}_B \left(\sum_{k=1}^{k-1} \mathbf{a}(\tilde{\omega}_{d,k}, \tilde{\omega}_{r,k})\tilde{x}_k + \mathbf{a}(\omega_{d,k}, \omega_{r,k})x_k + \boldsymbol{\epsilon} \right). \quad (20)$$

For brevity, define $\boldsymbol{\zeta} = \sum_{k=1}^{k-1} \mathbf{a}(\tilde{\omega}_{d,k}, \tilde{\omega}_{r,k})\tilde{x}_k \in \mathbb{C}^G$. The effects of these estimated targets are synthesized and treated as a known term.

For the k th target, the log-likelihood function is updated as

$$\begin{aligned} \mathcal{L}(\mathbf{y}; \boldsymbol{\theta}, x, \boldsymbol{\zeta}) = & \sum_{g=1}^G \left[\log \left(\Phi \left(\frac{u(\Re\{y_g\}) - \Re\{a_g(\boldsymbol{\theta})x + \boldsymbol{\zeta}_g\}}{\sigma/\sqrt{2}} \right) - \Phi \left(\frac{l(\Re\{y_g\}) - \Re\{a_g(\boldsymbol{\theta})x + \boldsymbol{\zeta}_g\}}{\sigma/\sqrt{2}} \right) \right) \right. \\ & \left. + \log \left(\Phi \left(\frac{u(\Im\{y_g\}) - \Im\{a_g(\boldsymbol{\theta})x + \boldsymbol{\zeta}_g\}}{\sigma/\sqrt{2}} \right) - \Phi \left(\frac{l(\Im\{y_g\}) - \Im\{a_g(\boldsymbol{\theta})x + \boldsymbol{\zeta}_g\}}{\sigma/\sqrt{2}} \right) \right) \right]. \end{aligned} \quad (21)$$

Fixing the parameters of all other targets, GNOMP updates the $\hat{\boldsymbol{\theta}}_k$ using the Newton step

$$\hat{\boldsymbol{\theta}}'_k = \hat{\boldsymbol{\theta}}_k - \left[\nabla^2 \mathcal{L}(\mathbf{y}; \hat{\boldsymbol{\theta}}_k, x, \boldsymbol{\zeta}) \right]^{-1} \Big|_{x=\hat{x}_{g,k}} \nabla \mathcal{L}(\mathbf{y}; \hat{\boldsymbol{\theta}}_k, x, \boldsymbol{\zeta}) \Big|_{x=\hat{x}_{g,k}}. \quad (22)$$

The refined amplitude $\hat{x}'_k$ can be further obtained via

$$\hat{x}'_k = \underset{x}{\operatorname{argmax}} \mathcal{L}(\mathbf{y}; \hat{\boldsymbol{\theta}}_k, x, \boldsymbol{\zeta}). \quad (23)$$

Once the k th target is optimized, the algorithm proceeds cyclically to update the others.

After all amplitude–frequency pairs have been refined, parameters of all k targets are collected into $\mathcal{P} = \{(\hat{x}'_k, \hat{\boldsymbol{\theta}}'_k), k = 1, \dots, K\}$. By fixing $\hat{\boldsymbol{\theta}}'$, the amplitudes are jointly updated by

$$\underset{\mathbf{x}}{\operatorname{maximize}} \mathcal{L}(\mathbf{y}; \{\hat{\boldsymbol{\theta}}'_k\}_{k=1}^K, \mathbf{x}). \quad (24)$$

The procedure is repeated until k reaches K , meaning that all existing K targets have been estimated.

IV. CRAMER–RAO BOUND

The MLE asymptotically approaches to the CRB. To evaluate the estimation performance of the 2D-GNOMP, the CRB is derived. For the one dimensional scenario, the CRB has been given in Lemma 1 [19]. For completeness, we present the CRB for the two dimensional case. Let $\boldsymbol{\kappa} \in \mathbb{R}^{4K}$ denote the set of all unknown deterministic parameters given as

$$\boldsymbol{\kappa} = [\boldsymbol{\omega}_r^T, \boldsymbol{\omega}_d^T, \Re^T\{\mathbf{x}\}, \Im^T\{\mathbf{x}\}]^T. \quad (25)$$

The Fisher information matrix is then expressed as [19]

$$\mathbf{I}(\boldsymbol{\kappa}) = \frac{2}{\sigma^2} \left[\frac{\partial \boldsymbol{\mu}(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}^T} \right]^T \boldsymbol{\Lambda} \left[\frac{\partial \boldsymbol{\mu}(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}^T} \right], \quad (26)$$

where

$$\boldsymbol{\mu}(\boldsymbol{\kappa}) = \begin{bmatrix} \Re\{\mathbf{A}(\omega_d, \omega_r)\mathbf{x}\} \\ \Im\{\mathbf{A}(\omega_d, \omega_r)\mathbf{x}\} \end{bmatrix} \in \mathbb{R}^{2G}, \quad (27)$$

and $\frac{\partial \boldsymbol{\mu}}{\partial \boldsymbol{\kappa}^T} \in \mathbb{R}^{2G \times 4K}$ is

$$\frac{\partial \boldsymbol{\mu}}{\partial \boldsymbol{\kappa}^T} = \begin{bmatrix} \frac{\partial \Re\{\mathbf{A}\mathbf{x}\}}{\partial \omega_r} & \frac{\partial \Re\{\mathbf{A}\mathbf{x}\}}{\partial \omega_d} & \Re\{\mathbf{A}\} & -\Im\{\mathbf{A}\} \\ \frac{\partial \Im\{\mathbf{A}\mathbf{x}\}}{\partial \omega_r} & \frac{\partial \Im\{\mathbf{A}\mathbf{x}\}}{\partial \omega_d} & \Im\{\mathbf{A}\} & \Re\{\mathbf{A}\} \end{bmatrix}. \quad (28)$$

$\boldsymbol{\Lambda}$ is a diagonal matrix, whose (i, i) th element can be represented by

$$\Lambda_{i,i} = h_B(\mu_i(\boldsymbol{\kappa}), \sigma^2), \quad (29)$$

where $h_B(x, \sigma^2)$ is given by

$$h_B(x, \sigma^2) = \sum_{d=0}^{b-1} \frac{\left[\phi \left(\frac{\tau_{d+1}-x}{\sigma/\sqrt{2}} \right) - \phi \left(\frac{\tau_d-x}{\sigma/\sqrt{2}} \right) \right]^2}{\Phi \left(\frac{\tau_{d+1}-x}{\sigma/\sqrt{2}} \right) - \Phi \left(\frac{\tau_d-x}{\sigma/\sqrt{2}} \right)}. \quad (30)$$

The CRB for the unknown parameters is given by the inverse of the Fisher information matrix, i.e.,

$$\text{CRB}(\boldsymbol{\kappa}) = \mathbf{I}^{-1}(\boldsymbol{\kappa}). \quad (31)$$

V. NUMERICAL SIMULATION

This section evaluates the estimation performance of 2D-GNOMP. The noise variance is set as $\sigma^2 = 1$. Based on the discussions in [20]–[23], the maximum full-scale range γ is calculated as $\max(|x_1|, \dots, |x_k|, \dots, |x_K|, 3\sigma/\sqrt{2})$, and the corresponding quantization interval is given by $\frac{2\gamma}{b}$, where $b = 2^B$ and B denotes the bit depth of the quantizer. For unquantized measurements, 2D-MNOMP [24] is employed and denoted as 2D-GNOMP ($B = \infty$).

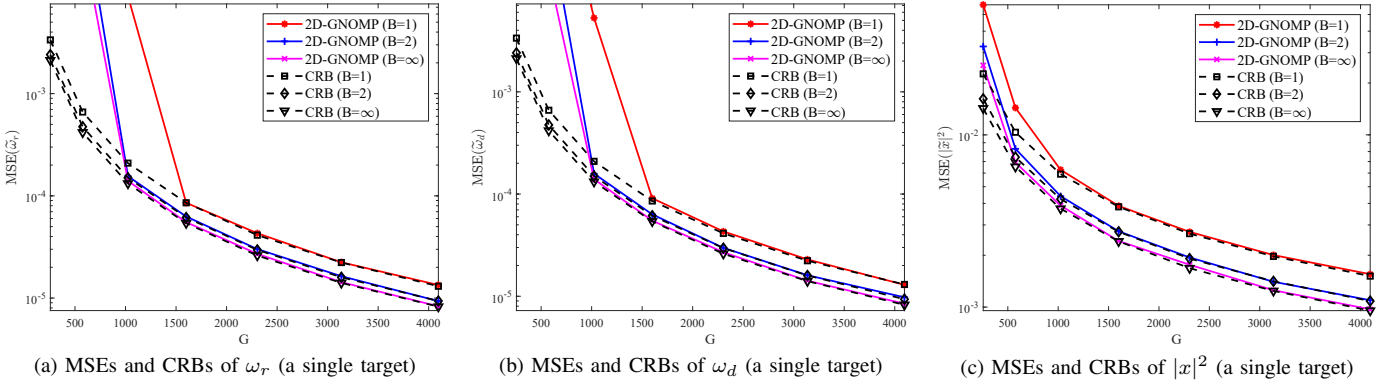


Fig. 1. MSE and CRB of frequencies and amplitude estimation versus the number of measurements N and M , averaged over 5000 Monte Carlo (MC) trials under different quantization resolutions, for the single target scenario.

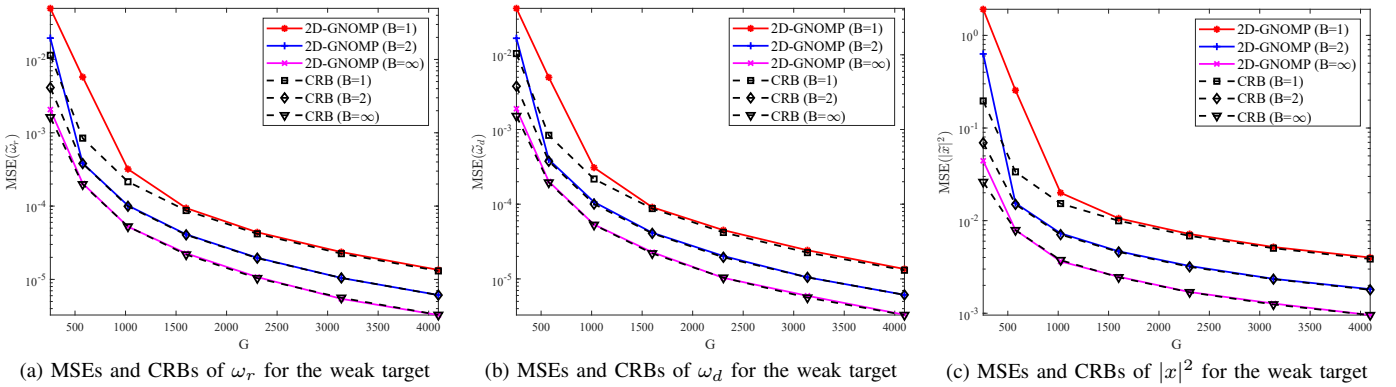


Fig. 2. MSE and CRB of frequencies and amplitude estimation versus the number of measurements N and M , averaged over 5000 MC trials under different quantization resolutions, for the coexisting strong and weak target scenario.

A. A Single Target Scenario

For the initial numerical simulations, we consider a scenario with a single target. The complex amplitude of the target signal is $-0.14 - 0.15j$, corresponding to a time-domain SNR of -13.6 dB, with range and Doppler angular frequencies of 0.34 and 1.26 , respectively. The two dimensions N and M are initially set to 16 and each increased in steps of 8 simultaneously. The mean squared errors (MSEs) of the frequency and amplitude estimates, together with the corresponding CRBs, are presented in Fig. 1.

It is observed that, as N and M increase, the MSEs approach the corresponding CRBs for 1 bit, 2 bit and unquantized cases. Among them, the unquantized case approaches to the CRB first, followed by the 2 bit quantization and 1 bit quantization. Once attaining to the CRB, the MSE of GNOMP under 2 bit quantization are close to that of the unquantized case, in terms of the frequency and amplitude estimation errors.

B. A Coexisting Strong and Weak Target Scenario

We further consider a scenario with coexisting strong and weak targets. The complex amplitudes of the targets are $\mathbf{x} = [0.05 + 0.33j, 0.06 - 1.86j]^T$, corresponding to time-

domain SNRs of -10 dB and 5 dB, respectively. The range and Doppler angular frequencies of the targets are $\omega_r = [0.34, 0.58]^T$ and $\omega_d = [1.26, 1.48]^T$. The MSEs of the frequency and amplitude estimates for the weak target, along with the corresponding CRBs, as shown in Fig. 2.

It can be seen that, with the strong and weak targets differing by 15 dB, the proposed 2D-GNOMP algorithm still attains MSEs closely matching the corresponding CRBs, indicating a high estimation accuracy. This result suggests that 2D-GNOMP can effectively mitigate the interference leakage from the strong target, which is critical for reliable weak target estimation in multi-target scenarios.

VI. CONCLUSION

This paper develops a 2D-GNOMP algorithm for super-resolution off-grid joint range-Doppler estimation in FMCW millimeter-wave radar under few-bit quantization, where conventional high-resolution and unquantized methods are not directly applicable, and derives the corresponding CRB as a benchmark. Numerical results in both single-target and strong-weak target scenarios show that the proposed method asymptotically approaches the CRB, indicating that near-optimal range-Doppler estimation remains achievable with few-bit ADCs.

REFERENCES

- [1] J. Park and C. Nguyen, "A new millimeter-wave step-frequency radar sensor for distance measurement," *IEEE Microw. Wireless Compon. Lett.*, vol. 12, no. 6, pp. 221–222, 2002.
- [2] W. Zhang, H. Li, G. Sun, and Z. He, "Enhanced detection of Doppler-spread targets for FMCW radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 4, pp. 2066–2078, 2019.
- [3] T. Kazaz, K. Jayachandra, A. Koppellar, and Y. Lu, "Joint multiple FMCW chirp sequence processing for velocity estimation and ambiguity resolving," in *Proc. IEEE Int. Radar Conf. (RADAR)*, pp. 1–6, 2023.
- [4] F. Engels, P. Heidenreich, M. Wintermantel, L. Stacker, M. Al Kadi, and A. M. Zoubir, "Automotive radar signal processing: Research directions and practical challenges," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 4, pp. 865–878, 2021.
- [5] Q. Xu, C. Zhao, F. Ding, Z. Chen, S. Wu, and W. Chen, "Ocean wave measurement using 77-GHz FMCW MIMO radar at low incidence angles," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
- [6] H. Cui and N. Dahnoun, "High precision human detection and tracking using millimeter-wave radars," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 36, no. 1, pp. 22–32, 2021.
- [7] K. V. Mishra, M. R. B. Shankar, V. Koivunen, B. Ottersten, and S. A. Vorobyov, "Toward millimeter-wave joint radar communications: A signal processing perspective," *IEEE Signal Process. Mag.*, vol. 36, no. 5, pp. 100–114, 2019.
- [8] P. Kumari, A. Mezghani, and R. W. Heath, "A low-resolution ADC proof-of-concept development for a fully-digital millimeter-wave joint communication-radar," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, pp. 8619–8623, 2020.
- [9] N. I. Bernardo, J. Zhu, Y. C. Eldar, and J. Evans, "Capacity bounds for one-bit MIMO Gaussian channels with analog combining," *IEEE Trans. Commun.*, vol. 70, no. 11, pp. 7224–7239, Nov. 2022.
- [10] A. Gupta, U. Madhow, and A. Arbabian, "Super-resolution in position and velocity estimation for short-range MM-wave radar," in *Proc. Asilomar Conf. Signals, Syst., Comput.*, pp. 1144–1148, 2016.
- [11] O. Bar-Shalom and A. J. Weiss, "DOA estimation using one-bit quantized measurements," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 3, pp. 868–884, 2002.
- [12] Y. Cheng, J. Karlsson, and J. Li, "Cram r-Rao bound for signal parameter estimation from modulo ADC generated data," *IEEE Trans. Signal Process.*, vol. 72, pp. 4268–4285, 2024.
- [13] X. Li, Z. Shi, L. Huang, Y. Xiao, and H. C. So, "Robust single-bit direction-of-arrival estimation with magnitude recovery," *IEEE J. Sel. Topics Signal Process.*, vol. 19, no. 6, pp. 943–958, 2025.
- [14] Y. Xiao, X. Li, T. Long, C. Qian, L. Huang, and H. C. So, "One-bit matched filter for direction-of-arrival estimation," *IEEE Trans. Veh. Technol.*, pp. 1–6, early access, 2025.
- [15] S. Hu, Y. Xiao, P. Wu, S. Liu, and L. Huang, "One-bit distributed radar target detection with range-Doppler estimation," in *Proc. 7th Int. Conf. Inf. Commun. Signal Process. (ICICSP)*, pp. 513–518, 2024.
- [16] F. Liu, H. Zhu, C. Li, J. Li, P. Wang, and P. V. Orlik, "Angular-domain channel estimation for one-bit massive MIMO systems: Performance bounds and algorithms," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2928–2942, Mar. 2020.
- [17] J. Zhu, H. Zhang, N. Zhang, J. Fang, and F. Qu, "Line spectrum estimation and detection with few-bit ADCs: Theoretical analysis and generalized NOMP algorithm," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 1, pp. 1–18, 2025.
- [18] N. Shlezinger, Y. C. Eldar, and M. R. D. Rodrigues, "Hardware-limited task-based quantization," *IEEE Trans. Signal Process.*, vol. 67, no. 20, pp. 5223–5238, 2019.
- [19] N. Zhang, J. Zhu, and Z. Xu, "Gridless multisnapshot variational line spectral estimation from coarsely quantized samples," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 3, pp. 2979–2993, 2023.
- [20] M. Xu, J. Zhu, J. Fang, N. Zhang, and Z. Xu, "CFAR based NOMP for line spectral estimation and detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 6971–6990, 2023.
- [21] Y. Cheng, X. Shang, and P. Stoica, "Interval design for signal parameter estimation from quantized data," *IEEE Trans. Signal Process.*, vol. 70, pp. 6011–6020, 2022.
- [22] G. Wang, J. Zhu, and Z. Xu, "Asymptotically optimal one-bit quantizer design for weak-signal detection in generalized Gaussian noise and lossy binary communication channel," *Signal Process.*, vol. 154, pp. 207–216, 2018.
- [23] J. Zhu, X. Lin, R. S. Blum, and Y. Gu, "Parameter estimation from quantized observations in multiplicative noise environments," *IEEE Trans. Signal Process.*, vol. 63, no. 15, pp. 4037–4050, 2015.
- [24] L. Han, X. Liu, N. Zhang, S. Wu, J. Zhu, and Z. Xu, "Two-dimensional multi-snapshot Newtonized orthogonal matching pursuit for DOA estimation," *Digit. Signal Process.*, vol. 121, Art. no. 103313, 2022.